



Today's agenda

↳ Max Subsequence Sum

↳ 0/1 Knapsack

↳ unbounded Knapsack

↳ iterative DP



AlgoPrep



Q) Max Subsequence Sum

↳ Given $arr[N]$ Calculate max subseqⁿ sum.

Note: Subseqⁿ with 2 adj element is not allowed.

$arr[3]: \{9, 14, 3\} : ans = 14$

$arr[4]: \{9, 14, 13, 24\} : ans = 38$

$arr[3]: \{13, 14, 2\} : ans = 15$

$arr[4]: \{-4, -3, -2, -10\} : ans = 0$

Idea 1 → incorrect idea

↳ max { Sum of every even index element, Sum of every odd index element }

$arr[4]: \{ \overset{0}{9}, \overset{1}{1}, \overset{2}{13}, \overset{3}{24} \} : ans = 33$

Idea 2

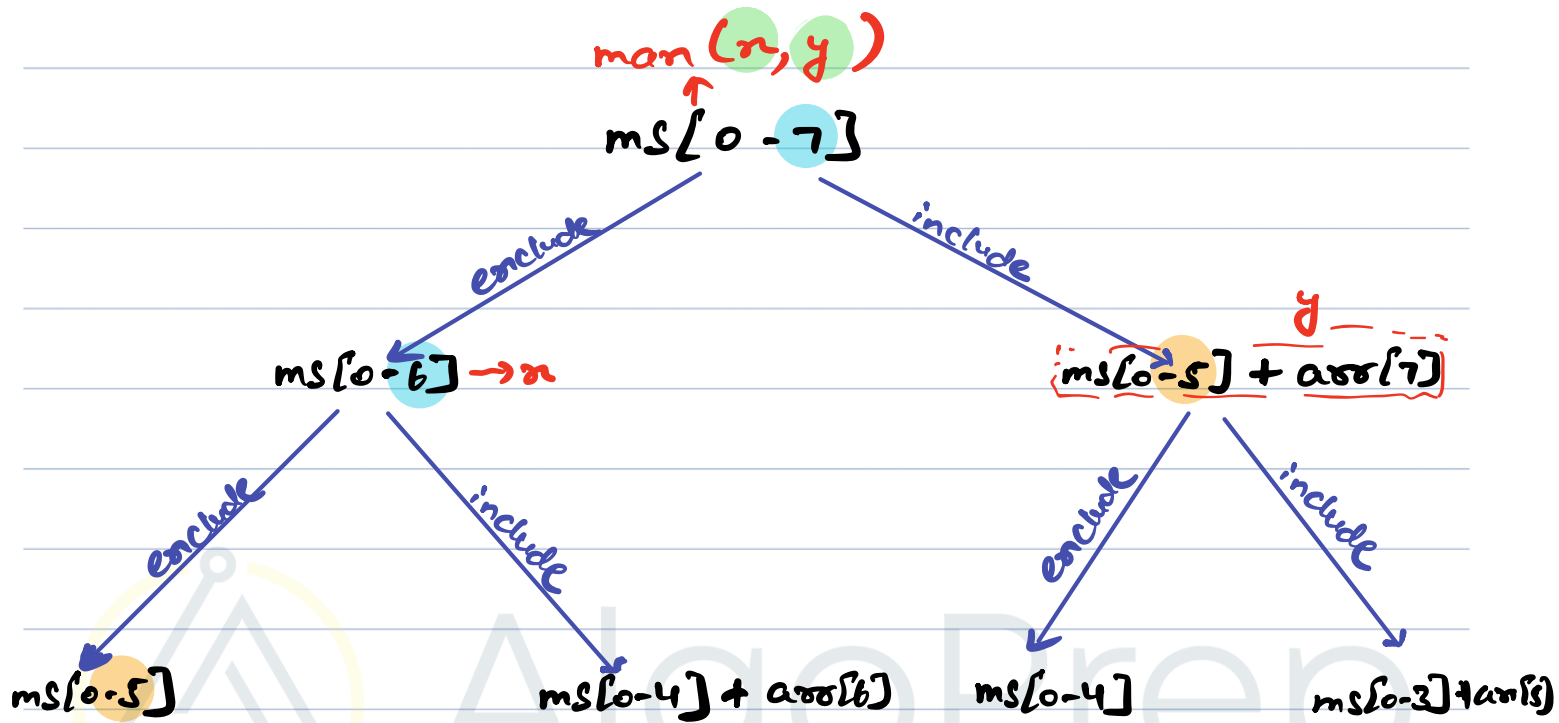
↳ for every subsequenceⁿ (following the note), find the sum and pick the max.

T.C: $O(2^N * N)$ S.C: ...



Slide 3

arr[8]: { 2⁰ -1¹ -4² 5³ 3⁴ -1⁵ 4⁶ 2⁷ }





// Pseudo code

```
int dp[n] = {-1}
```

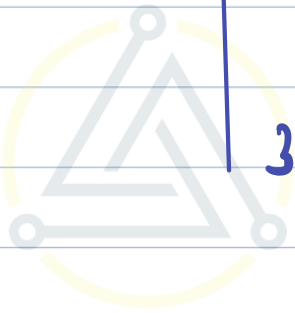
```
int maxSubseq (int arr[], int i) {  
    if (i < 0) { return 0; }  
    if (dp[i] != -1) { return dp[i]; }
```

T.C: $O(n)$

S.C: $O(n)$

```
    int x = maxSubseq (arr, i-1);  
    int y = maxSubseq (arr, i-2) + arr[i];
```

```
    dp[i] = max(x, y);  
    return dp[i];  
}
```



AlgoPrep



Q) 0-1 Knapsack

↳ Given n items each with a weight and value, find max value that can be obtained by picking items such that total weight of all items $\leq K$.^{given}

Note: {1 item can be picked once}.

Ex: $n=4$ $K=50$
wt: 20 10 30 40 $\rightarrow$ ans = 220
value: 100 60 120 150

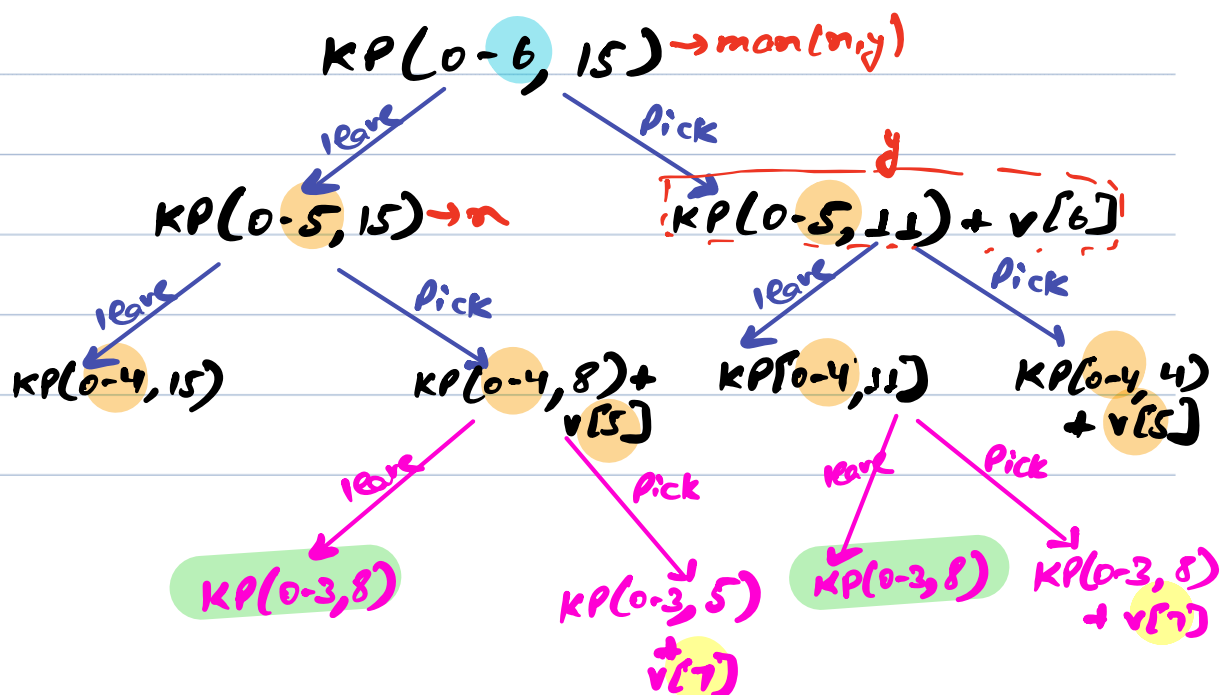
Idea 1

↳ generate all subsequence of items and pick the one with total weight $\leq K$ and max val.

T.C: $O(2^n)$

Idea 2

$K=15$
wt: 3 1 5 4 3 7 4
v: 2 2 8 3 7 10 5





//Pseudo code

```
int dp[W][K+1] = {-1};
```

```
int knapsack (int w[], int v[], int k, int i){
```

```
    if (i == -1 || k == 0) { return 0; }
```

```
    if (dp[i][k] != -1) { return dp[i][k]; }
```

```
    int n = knapsack(w, v, k, i-1); //include
```

```
    int y = 0;
```

```
    if (k >= w[i]) { //include
```

```
        y = knapsack(w, v, k-w[i], i-1) + v[i];
```

```
    }
```

```
    dp[i][k] = max(n, y);
```

```
    return max(n, y);
```

```
}
```

T.C: $O(W \times K)$

S.C: $O(W \times K)$



Q) unbounded Knapsack

↳ Exactly same as above problem.

Note: A single item can be picked as many times as possible.

Ex: $N=4$ $K=50$

	0	1	2	3	
wt[]:	20	10	30	40	
value:	100	60	120	150	

→ ans: 300

UKP(0-3, 50)

leave

UKP(0-2, 50)

Pick

UKP(0-3, 10) + v[3]



//Pseudo Code

```
int dp[N][K+1] = {-1};
```

```
int knapsack (int w[], int v[], int K, int i) {  
    if (i == -1 || K == 0) { return 0; }  
    if (dp[i][K] != -1) { return dp[i][K]; }
```

```
    int n = knapsack(w, v, K, i-1); //include
```

```
    int y = 0;
```

```
    if (K >= w[i]) { //include
```

```
        y = knapsack(w, v, K-w[i], i) + v[i];
```

```
    }
```

```
    dp[i][K] = max(n, y);
```

```
    return max(n, y);
```

```
}
```

T.C: $O(W \times K)$

S.C: $O(W \times K)$

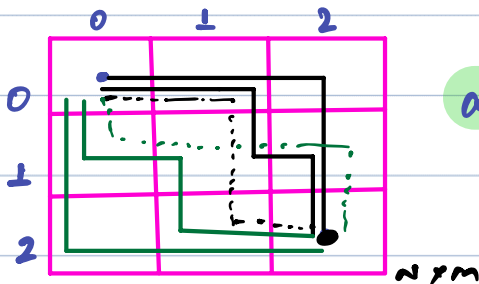


Iterative DP

Q) Number of ways to go from $(0,0) \rightarrow (n-1, m-1)$ cell.

Movement allowed: cell $\xrightarrow{\text{1 step right}}$ or $\downarrow \text{1 step bottom}$

Ex:



ans: 6

RRDD

DDRR

RDRD

DRDR

RDDR

DRRD

$PC(i, j) \rightarrow PC(i-1, j) + PC(i, j-1)$

	0	1	2
0	1	1	1
1	1	2	3
2	1	3	6



Q) Minimum Path Sum

↳ Given a 2d array, filled with non-negative numbers, find a Path from $(0,0) \rightarrow (n-1, m-1)$ which minimizes the total cost Path.

Movement allowed: cell $\xrightarrow[\text{right}]{\text{1 Step}}$ or $\downarrow \text{1 Step Bottom}$
 R D

Ex:

	0	1	2
0	2	3	5
1	1	6	4
2	9	2	7

	0	1	2
0	2	5	10
1	3	9	13
2	12	11	18



Q) 0-1 Knapsack

Given n items each with a weight and value, find max value that can be obtained by picking items such that total weight of all items $\leq K$.^{given}

Note: {1 item can be picked once}.

$n=7$

	0	1	2	3
$w[i]$:	3	1	5	4
$v[i]$:	2	2	8	3

weight (K)

dp:

	0	1	2	3	4	5	6	7
0	0	0	0	2	2	2	2	2
1	0	2	2	—	—	—	—	—
2	0	—	—	—	—	—	—	—
3	0	—	—	—	—	—	—	—

$KP(0-6, 15) \rightarrow \text{max}(n, y)$

leave
 $KP(0-5, 15) \rightarrow n$

Pick
 $KP(0-5, 11) + v[6]$

$KP(0-1, 1)$

leave
 $KP(0-0, 1) \rightarrow 0$

Pick
 $KP(0-0, 0) + 2$

$KP(0-1, 2)$

leave
 $KP(0-0, 2) \rightarrow 0$

Pick
 $KP(0-0, 1) + 2$

